

Investigating the anchorage performance of full-grouted anchor bolts with a modified numerical simulation method

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ABSTRACT

To reveal the failure mechanism of full-grouted bolting system, revised Pile elements were selected to numerically reveal the anchoring performance of full-grouted anchor bolts. A tri-linear bond-slip model was embedded in the Pile elements. A three-dimensional mining stope was created. A rectangular main roadway reinforced with full-grouted anchor bolts was simulated. Then, the retreat mining method was used to exploit the coal seam. The impact of buried depth, pillar size and shearing strength of the grout/rock contact surface was studied. It showed that the coal seam's burial depth significantly influenced the anchorage performance of the anchor bolts. Moreover, increasing the shearing strength of the grout/rock contact surface was conducive to improving the anchorage performance of anchor bolts and preventing roadway convergence. Simulation results were compared against experimental laboratory test and in-situ observation outcome. It was conducive to revealing the anchorage mechanism of full-grouted anchor bolts under the in-situ condition.

1. Introduction

Underground coal mining plays a significant role in energy supply [1]. For example, in China, the statics data shows that more than 60 % of coal resources are supplied with underground coal mining methods [2]. Therefore, keeping the safety and normal running of underground coal mines is vital in guaranteeing the stability of the current society [3].

To keep the safety of underground coal mines, improving the ground control methods and techniques is always the research hotspot [4–6]. Among them, using full-grouted anchor bolts to reinforce jointed rock masses is a fundamental part [7–9].

After installation of anchor bolts, deformation of rock masses induces the elongation of anchor bolts. This elongation leads to tensile loading of anchor bolts. Consequently, the load is transferred between rock masses and anchor bolts. In-situ observation proves that insufficient load transfer between rock masses and bolts leads to bed separation, which may finally result in roof collapsing disasters [10]. For example, it is reported that in the open-off cut labelled 9-22140 in a colliery, a serious roof collapsing accident occurred, in which the collapsed roof had a height of 6.5 m [11]. Moreover, the reinforced cable bolts whose length was 6.5 m were completely

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Nomenclature

s	the shearing stress (Pa);
d	the interfacial deformation (m);
σ_{zz}	the in-situ vertical stress (Pa);
γ	the rock's volume weight (N/m^3);
H	the buried depth (m);
σ_b	the uniaxial stress in the anchor bolt (Pa);
F_b	the uniaxial force in the anchor bolt (N);
A	the area of the anchor bolt (m^2).

pulled out with the roof.

Investigating the loading mechanics of anchor bolts is highly significant in maintaining the safety of underground openings [12,13]. To reveal the loading mechanics of anchor bolts, performing experimental tests is a relatively better choice. However, performing experiments require the test facility, labor work and financial support.

Compared with that, performing numerical simulation has less requirement on the platform. Moreover, users can adjust the numerical simulation algorithm based on secondary development methods. Consequently, numerical simulation becomes more widely accepted in analyzing the anchorage performance of anchor bolts [14].

Ma, *et al.* [15] put forward a numerical method to study the anchorage performance of anchor bolts. Kang, *et al.* [16] studied the anchorage performance of resin bolts. Bahrani and Hadjigeorgiou [17] indicated that the so-called cable element in the UDEC software underestimated the shearing capacity of anchor bolts. Chang, *et al.* [18] indicated that increasing the shearing strength of the grout annuls led to changing of failing types. Shang, *et al.* [19] created a three-dimensional discrete element algorithm to study the failing type of the rock reinforcement system.

Nie, *et al.* [20] indicated that the reinforcement rock unit induced by a single anchor bolt was different from a group of bolts. Deb and Gujjala [21] developed an enriched finite element simulation method to analyze the interaction between grouted anchor bolts and joints in rock masses. Yan, *et al.* [22] put forward a numerical approach to simulate the tensile rupture of resin grouted bolts. Nie, *et al.* [23] indicated that the bonding and debonding process of anchor bolts can be divided into three stages.

Yokota, *et al.* [24] studied the contact behavior between anchor bolts and ambient rock mass based on the discontinuous deformation analysis. Zuo, *et al.* [25] put forward a numerical method to reveal the matching relationship between anchor bolts and the drilled borehole. Ho, *et al.* [26] numerically studied the failure process of the anchor bolts. Singh, *et al.* [27] studied the stress state of grouted anchor bolts subjected to pulling and shearing condition.

Fu, *et al.* [28] designed the orientation and lifting instrument for resin bolts and numerically revealed the working principal of this instrument. Mohamed, *et al.* [29] analyzed the anchorage performance of bolts reinforced in coal ribs. Li, *et al.* [30] used the numerical approach to reveal the anchorage performance of anchor bolts reinforced in composite rock strata.

The above research played a marked role in disclosing the failure mechanism of anchor bolts. Among kinds of numerical simulation method, the structural element in the software of FLAC3D is more widely used. In the recent research, Chen and Li [31] embedded the tri-linear model in the Pile element to improve the simulation effect. In this context, the authors continued the previous work by simulating the anchorage performance of anchor bolts subjected to the retreat coal mining.

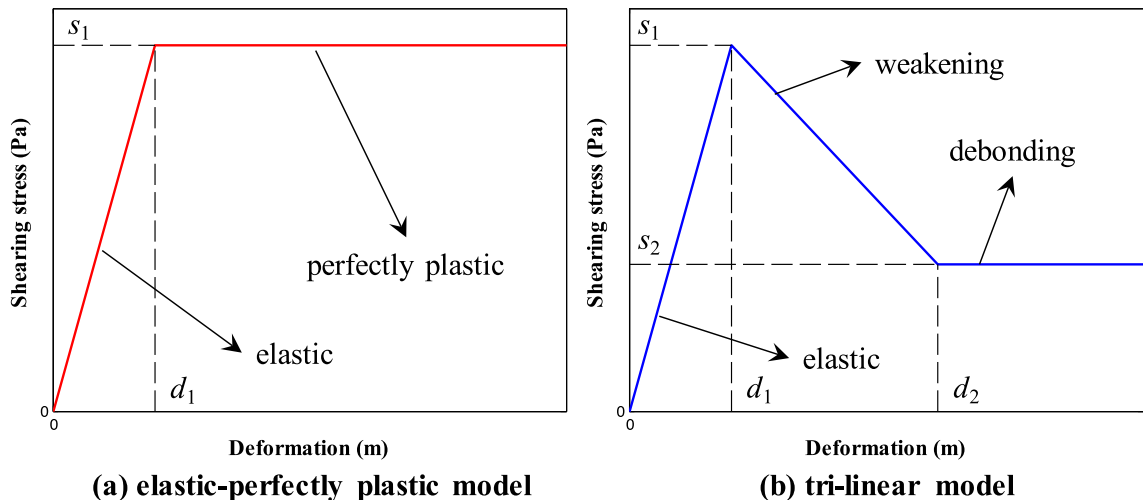


Fig. 1. Bonding and debonding behavior.

First, the revised simulation approach is described. Then, the numerical simulation scheme about the anchor bolts is illustrated. After that, the authors indicated the recommendation of the future work.

2. Description of the revised simulation approach

2.1. Bonding-debonding relationship

The bonding-debonding relationship describes the varying of the shearing stress at the anchorage contact surface with the deformation. It should be noted that this paper focused on the failure at the grout/rock contact face in the coal mining engineering. The reason is that when the threaded anchor bolts are used, the mechanical interlock and friction between the anchor bolt and grout are usually improved. Moreover, because of the poor adhesion at the borehole wall, slipping may occur at the grout/rock contact face [32]. In fact, in-situ experimental tests proved that when the anchorage tests were conducted under the weak rock environment, failure was more likely to occur at the grout/rock contact face [33,34].

A review of the previous research indicates that the tri-linear bond-slip model was initially developed by Aydan, *et al.* [35]. They used it to describe the failure process of the anchorage contact surface. Later, Benmokrane, *et al.* [36] performed experimental tensile tests on grouted anchor bolts, proving that this model was valid in describing the bonding and debonding performance of the anchorage contact surface.

Therefore, in this context, the tri-linear model was selected to substitute the elastic-perfectly plastic model of the Pile element in FLAC3D (Fig. 1). Apparently, the elastic-perfectly plastic model assumes that the shearing stress of the grout/rock contact surface remains constant after the shearing strength. Compared with that, the tri-linear model defines that after the shearing strength, the shearing stress decreases linearly with the deformation. Then, as the shearing stress decreases to the residual shearing strength, it remains constant. Therefore, the tri-linear model can successfully simulate the post-failure state of the grout/rock contact surface, as shown in Eqs. (1), (2) and (3).

$$s = \frac{s_1}{d_1} d \quad (1)$$

$$s = \frac{s_1 - s_2}{d_1 - d_2} s + \frac{s_2 d_1 - s_1 d_2}{d_1 - d_2} \quad (2)$$

$$s = s_2 \quad (3)$$

where s is the shearing stress (Pa); d is the interfacial deformation (m).

2.2. Embedding the revised model in the software

A numerical pulling test was established with FLAC3D to compare the loading state of anchor bolts when those two models shown in Fig. 1 were used. Embedding the tri-linear model in FLAC3D follows the same procedures described by the authors in the latest research [31].

In this sub-section, a threaded anchor bolt whose diameter was 25 mm was tested. The grouted length was 2 m and the bolt modulus was 200 GPa. In this numerical pulling test, a slow pulling velocity of 0.5 $\mu\text{m/s}$ was fixed on one end of the anchor bolt. Then, the

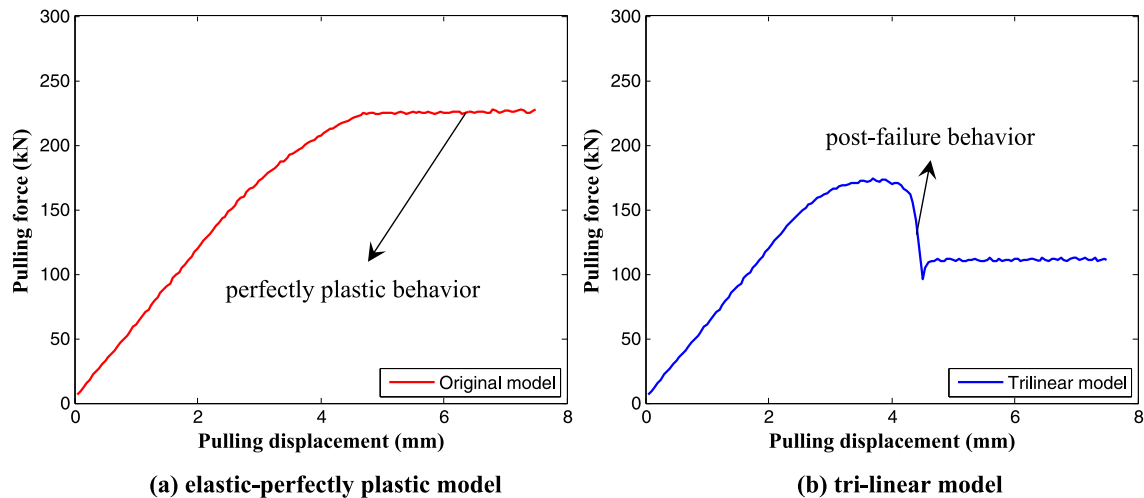


Fig. 2. Anchorage performance of anchor bolts.

tensile pulling capacity and displacement were recorded. As those two models were used, the comparison between the pulling outcome is shown in Fig. 2.

Apparently, there is a significant difference between the outcome. As the elastic-perfectly plastic model was used, after the pulling capacity reached the peak of 227.6 kN, it remained constant. The reason was that after the shear stress at the anchorage contact surface reached the maximum, it remained constant. It means when the shear stress reached the maximum, there was no declining of the shear stress. Consequently, for anchor bolts, when the pulling force reached the peak, it did not decline and remained constant. More importantly, there was no failure of the bolting system. Apparently, this was not a true reflection of the anchorage performance.

In contrast, as the tri-linear model was used, after the pulling capacity reached the peak of 173.7 kN, it dropped directly. More importantly, with the increasing pulling displacement, the pulling capacity rose again and remained at a residual capacity of around 110 kN.

Ma, et al. [37] performed uniaxial tests on anchor bolts in the laboratory to study the anchorage performance. The outcome shows that as the anchor bolt reached the maximum capacity, the bearing capacity dropped directly. Moreover, obvious residual bearing capacity occurred in the post-failure section because of the frictional effect (Fig. 3).

Through comparing the numerical simulation outcome and the experimental test outcome, the elastic-perfectly plastic model neglected the post-failure action of the anchorage contact surface. In contrast, the revised Pile element with the tri-linear model can better reflect the anchorage performance of anchor bolts, especially the post-failure action. Therefore, in the following section, the tri-linear model was used to analyze the anchorage performance of anchor bolts under the in-situ scenario.

3. Numerical simulation scheme

3.1. Three-dimensional mining stope creation

To simulate the anchorage performance of full-grouted anchor bolts, a three-dimensional digital mining stope was created (Fig. 4). The whole mining stope had a height of 55 m. As for the length, it was 200 m.

Along the vertical axis, this mining stope is divided into five different layers, including the roof, the main roof, the immediate roof, the coal seam and the floor (Fig. 5). The thickness of those layers ranged between 4 m and 28 m.

The rectangular main roadway was excavated in the coal seam. Its width is 5 m. As for its height, it is 4 m. At the roadway's left side, the coal pillar's length was 20 m. At the roadway's right side, a coal pillar with the size of 100 m was set.

3.2. Pre-defined setting

Considering that this mining stope was same along the axis of Y, to improve the calculating efficiency, a unit width of 1 m was finally simulated. The boundary condition of this mining stope was set as shown in Fig. 6. Specifically, the roller boundary was applied at four vertical sides of the mining stope. Moreover, at the lower side of the mining stope, the fixed boundary was set. Last, at the top side of the mining stope, a compensated stress of 7 MPa was applied.

For the layers of roof, main roof, immediate roof and floor, the Mohr-Coulomb model was set. Then, for the layer of the coal seam, the Strain-Softening model was defined. As for the input properties, they are tabulated in Table 1.

Under the gravitational compaction, this mining stope was solved automatically until the average unbalanced-force ratio reached the threshold of 1×10^{-5} . After that, the roadway was excavated. Meantime, installation of the 3 m long rock bolts and 7 m long cable bolts were performed. It should be noted that in mining engineering, there is a marked difference between rock bolts and cable bolts. First, the length of them is different. Rock bolts usually have a length less than 3 m. In contrast, cable bolts usually have a much longer length. Second, the manufacturing process is different. Rock bolts are usually cylindrical smooth bars or threaded bars. However, cable

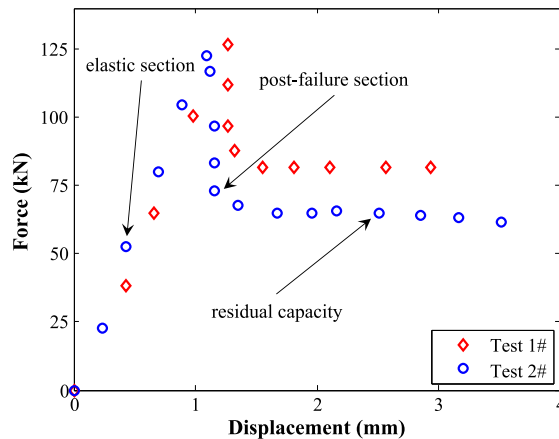


Fig. 3. Laboratory test outcome of anchor bolt [37].

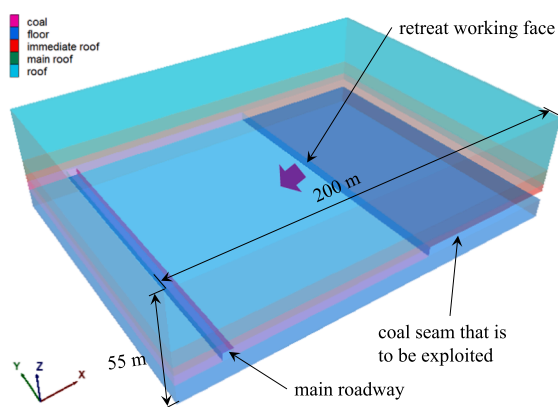


Fig. 4. The three-dimensional digital mining stope.

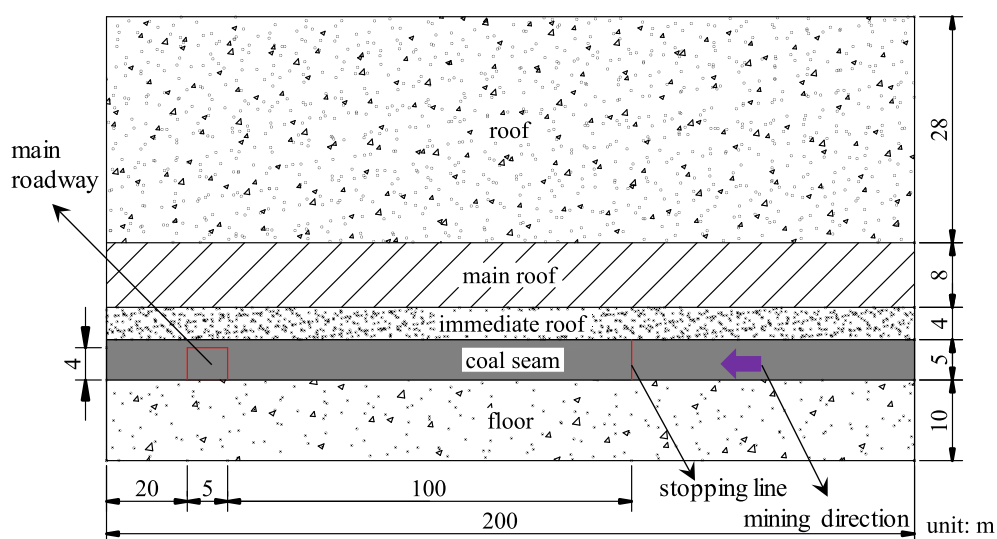


Fig. 5. Layout of the rock strata layers along the vertical axis.

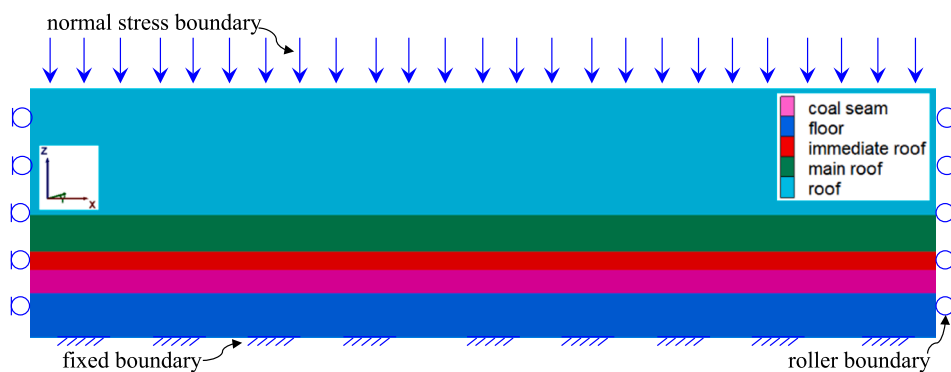


Fig. 6. Boundary condition for the numerical mining stope.

bolts are flexible tendons consisting of wires. The anchor bolt installation layout is shown in Fig. 7. All bolts were fully grouted.

Materials of those bolts are tabulated in Table 2.

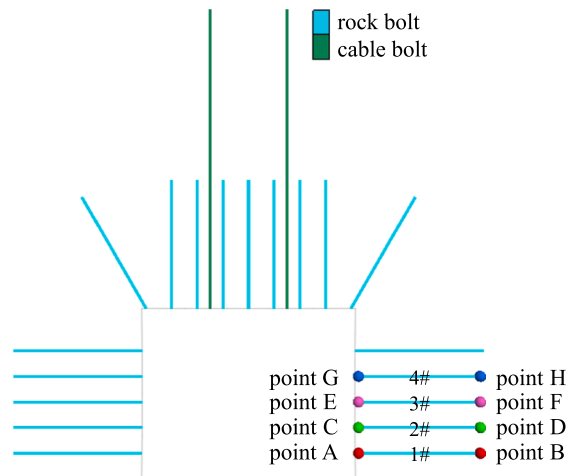
For the grout/rock contact surface, the corresponding input relationship is tabulated in Table 3.

After that, the retreat working face started mining towards the roadway. In FLAC3D, the null model was used to simulate this

Table 1

Input properties of each layer in the mining stope.

Layer name	Unit	Roof	Main roof	Immediate roof	Coal seam	Floor
Young's modulus	GPa	12	10.5	9.5	6	11
Poisson's ratio	–	0.25	0.23	0.15	0.22	0.24
Cohesion	MPa	5	3.5	3	2	3
Internal friction angle	°	33	32	29	28	32
Tension	MPa	3	2.5	1.5	1.3	2.2
Density	kg/m ³	2700	2500	2400	1400	2300

**Fig. 7.** Installation of anchor bolts around the roadway.**Table 2**

The bolt materials in this simulation.

Type	Unit	Rock bolt	Cable bolt
Bolt diameter	mm	25	32
Borehole perimeter	mm	87.96	109.96
Sectional area of anchor bolts	mm ²	490.9	804.2
Young's modulus	GPa	200	220
Poisson's ratio	–	0.25	0.25
Rupture capacity	kN	260	320

retreat mining process. After the retreat working face stopped at the stopping line, the numerical mining stope was calculated until the average unbalanced-force ratio dropped to 1×10^{-5} again. After the numerical mining stope equilibrated, the tensile capacity in anchor bolts and the shearing stress at the grout/rock contact surface was analyzed.

3.3. Investigated parameters

Based on the above-mentioned simulation process, the impact of three different factors on the anchorage performance of anchor bolts was studied: buried depth, pillar size and shearing strength of the grout/rock contact surface.

3.3.1. Buried depth

Considering that the coal seam may be buried at different depths dependent on the geological condition, this paper first studied the

Table 3

Input relationship of the grout/rock contact surface.

Type	unit	grout/rock contact surface
Shearing strength	MPa	2.5
Deformation where shearing strength occurred	mm	5
Residual shearing strength	MPa	0.5
Deformation where residual shearing strength occurred	mm	8

impact of buried depth on the anchorage performance of anchor bolts.

The in-situ perpendicular stress in the coal seam has a direct relationship with the buried depth, as shown in Eq. (4):

$$\sigma_{zz} = \gamma H \quad (4)$$

where σ_{zz} is the in-situ vertical stress (Pa); γ is the rock's volume weight (N/m^3); H is the buried depth (m).

Based on this concept, to simulate the different buried depths, the compensated stress at the top of this numerical mining stope was changed. Specifically, three different compensated stresses were applied at the top of this numerical mining stope: 3 MPa, 7 MPa and 10 MPa.

For those three compensated stresses, as this numerical mining stope initially equilibrated, the vertical stress in the coal seam was around 4 MPa, 8 MPa and 11 MPa (Fig. 8). Substituting the obtained vertical stress into Eq. (1) and assuming that the average volume weight of the rock strata was 27 kN/m^3 , the corresponding buried depths of the coal level were 148 m, 296 m and 407 m.

Then, the retreat mining process was performed. The working face stopped at the stopping line and the corresponding pillar size was 100 m. After the numerical mining stope equilibrated, the failure state of the numerical mining stope is shown in Fig. 9. Apparently, the buried depth made a difference on the failing type of the rock masses around the roadway. As the coal seam's buried depth was 148 m, with the current bolting method, there was little failure around the roadway. However, as the buried depth ascended to 296 m, apparent shear failure occurred in left side and right side of roadway. Moreover, tension failure occurred in the roadway's roof and floor. With the coal seam's burial depth increasing to 407 m, the current bolting method could not guarantee the roadway's stability. The volume of the failed rock masses around the roadway significantly ascended.

Meantime, the tensile force state in the anchor bolts is plotted (Fig. 10). Apparently, the buried depth had an apparent effect on the tensile force state in anchor bolts. As the buried depth was 148 m, the maximum tensile capacity occurred in the roof cable bolts. Moreover, it only had a small value of around 30.1 kN. As for the bolts fixed in two sides of the roadway, much less tensile force generated. With the increasing buried depth, the maximum tensile force in anchor bolts also ascended, ascending to around 72 kN. More interestingly, as the buried depth ascended to 407 m, the maximum tensile force occurred in rock bolts fixed in two sides of the roadway, which was around 136.6 kN. In this sub-section, the tensile force in the roof cable bolts was much smaller than the rock bolts that were fixed in two sides of the roadway.

To verify that the numerical simulation outcome was reliable, the uniaxial stress varying along the anchor bolt was compared with the in-situ observation outcome. The uniaxial stress in the anchor bolt was calculated with Eq. (5).

$$\sigma_b = \frac{F_b}{A} \quad (5)$$

where σ_b is the uniaxial stress in the anchor bolt (Pa); F_b is the uniaxial force in the anchor bolt (N); A is the area of the anchor bolt (m^2).

The anchor bolt 3# was regarded as an example and the corresponding uniaxial stress varying along the anchor bolt is illustrated with Fig. 11(a). Apparently, the uniaxial stress reached the peak in the middle part of the grouted length. When the compensated stress

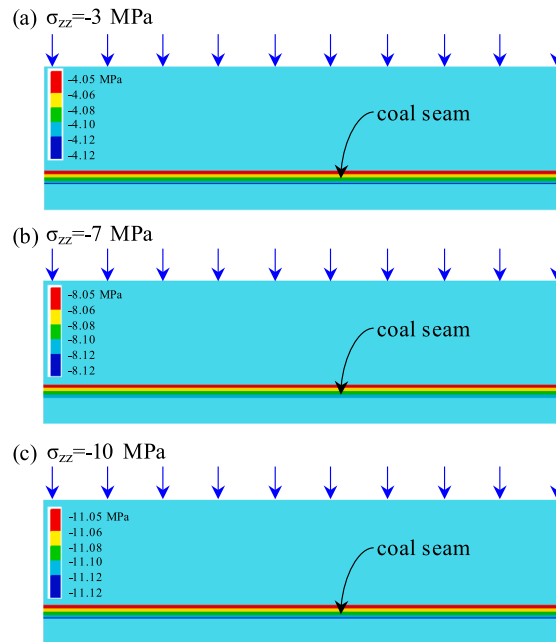


Fig. 8. Coal seam's vertical stress at the initial equilibrium: (a) compensated stress was 3 MPa; (b) compensated stress was 7 MPa; (c) compensated stress was 10 MPa.

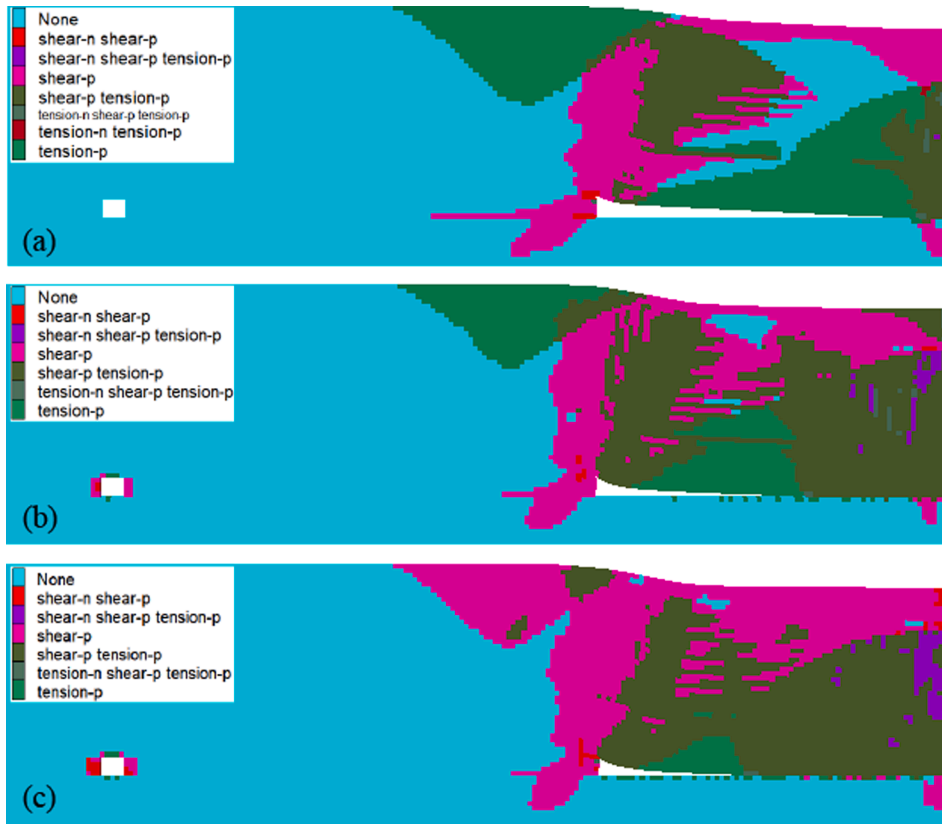


Fig. 9. Plastic zone state: (a) compensated stress was 3 MPa; (b) compensated stress was 7 MPa; (c) compensated stress was 10 MPa.

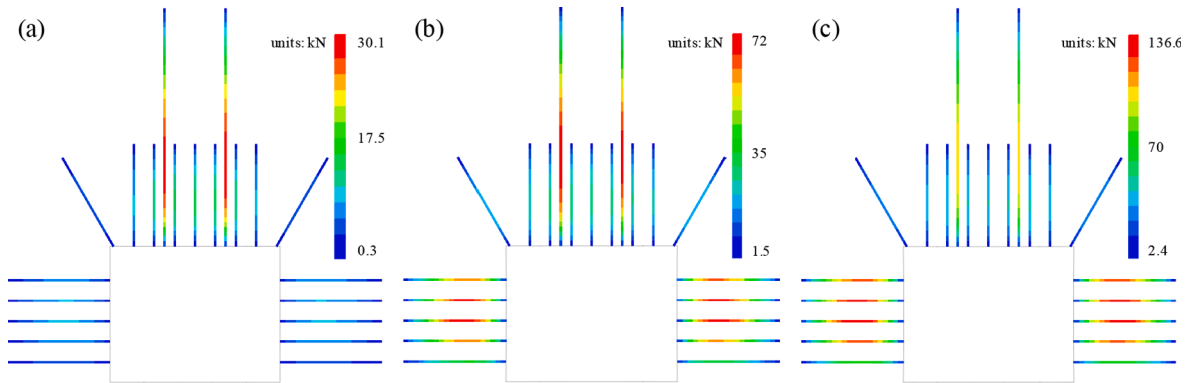


Fig. 10. Tensile force state along anchor bolts: (a) compensated stress was 3 MPa; (b) compensated stress was 7 MPa; (c) compensated stress was 10 MPa.

was 10 MPa, the maximum uniaxial stress in the anchor bolt was 273.5 MPa. Then, it declined towards two terminals of the anchor bolt. This declining pattern was independent on the stress in the coal seam. Apparently, when the compensated stresses were 3 MPa and 7 MPa, this declining pattern was still valid. More importantly, Sun [38] performed in-situ observation on the uniaxial stress varying along the anchor bolt and the corresponding data is illustrated with Fig. 11(b). Apparently, in the field, the uniaxial stress in the anchor bolt reached the peak in the middle part and was minimal around two terminals of the anchor bolt. This agreement between the numerical outcome and the in-situ observation outcome further confirmed that the revised pile element was robust in studying the anchorage performance of anchor bolts.

Then, the shearing stress state at the grout/rock contact surface along the anchor bolts is plotted, as shown in Fig. 12. In this figure, positive values represent that the shearing stress at the grout/rock contact surface points towards the internal rock orientation. In contrast, negative values represent that the shearing stress at the grout/rock contact surface points towards the roadway excavation

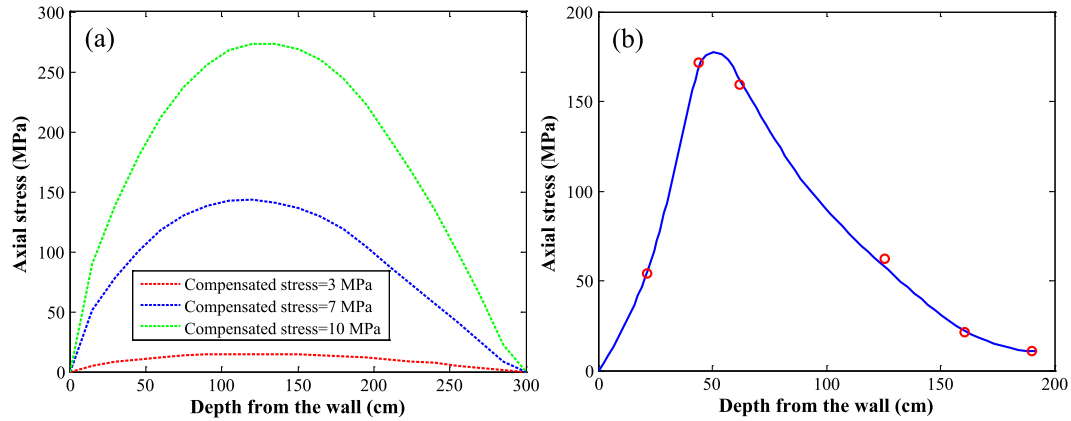


Fig. 11. Correlation between numerical and in-situ observation outcome regarding the uniaxial stress varying: (a) numerical simulation; (b) in-situ observation [38].

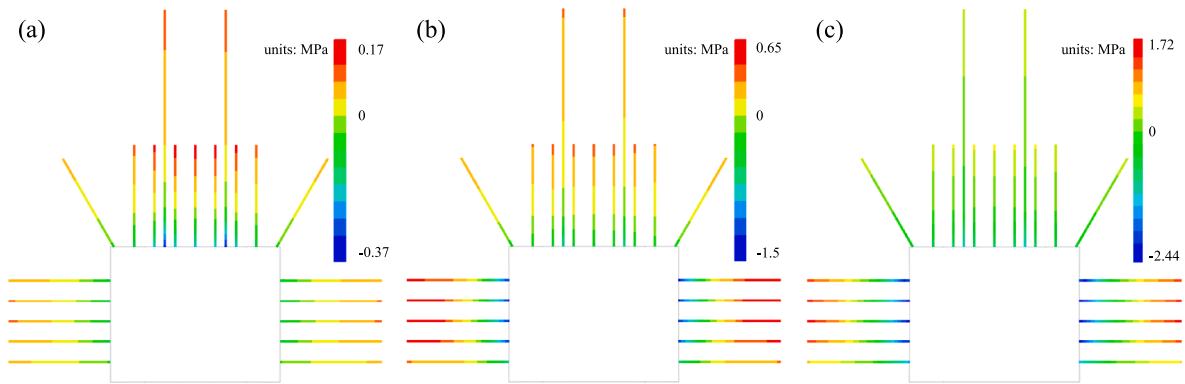


Fig. 12. Shearing stress state at the grout/rock contact surface: (a) compensated stress was 3 MPa; (b) compensated stress was 7 MPa; (c) compensated stress was 10 MPa.

orientation. The value of zero indicates the neutral point. It apparently shows that for anchor bolts, the interfacial shearing stress around the external part of the anchor bolts points towards the roadway. In contrast, the interfacial shearing stress around the internal part of the anchor bolts points towards the internal rock orientation. Moreover, neutral points occurred between two ends of anchor bolts. This finding was consistent with the in-situ observation outcome [38].

It indicated that the buried depth influenced the maximum shearing stress at the grout/rock contact surface. As the buried depth was 148 m, the maximum shearing stress at the grout/rock contact surface was only 0.17 MPa. As the buried depth ascended to 296 m, the maximum shearing stress at the grout/rock contact surface ascended to 0.65 MPa. As the buried depth ascended to 407 m, the maximum shearing stress at the grout/rock contact surface ascended to 2.44 MPa, which was around the shearing strength of the grout/rock contact surface. This indicated that with the ascending buried depth, the shearing stress of the grout/rock contact surface could entirely develop.

To further reveal the development process of the shearing stress at the grout/rock contact surface, the output relationships of 8 nodes shown in Fig. 7 were recorded. Specifically, the scenario in which the compensated stress was 10 MPa was used as an analysis case. The corresponding output results are shown in Fig. 13. Basically, the development process of the shearing stress at the grout/rock contact surface followed the pre-defined model. This further indicated that the model was successfully embedded in the Pile element in FLAC3D.

It also indicated that the development process of the shearing stress at the external part of the anchor bolts was different from the internal part. For the external part of anchor bolts, the shearing stress first ascended to the interfacial shearing strength, such as nodes C and G. After that, the shearing stress at those points started decreasing. In contrast, the shearing stress at the internal part of the anchor bolts always ascended, such as nodes B, D, F and H. Moreover, the shearing stress at the internal part of bolts did not reach the interfacial shearing strength.

As for the reason for this difference, it was deferred that in the external part of anchor bolts, convergence of the rock masses was relatively larger. This led to the larger deformation of the grout/rock contact surface. Following the pre-defined model, the shearing stress of the grout/rock contact surface encountered the elastic and plastic action successively. Compared with that, in the internal part of anchor bolts, convergence of the rock masses was much smaller. This led to the result that the deformation of the

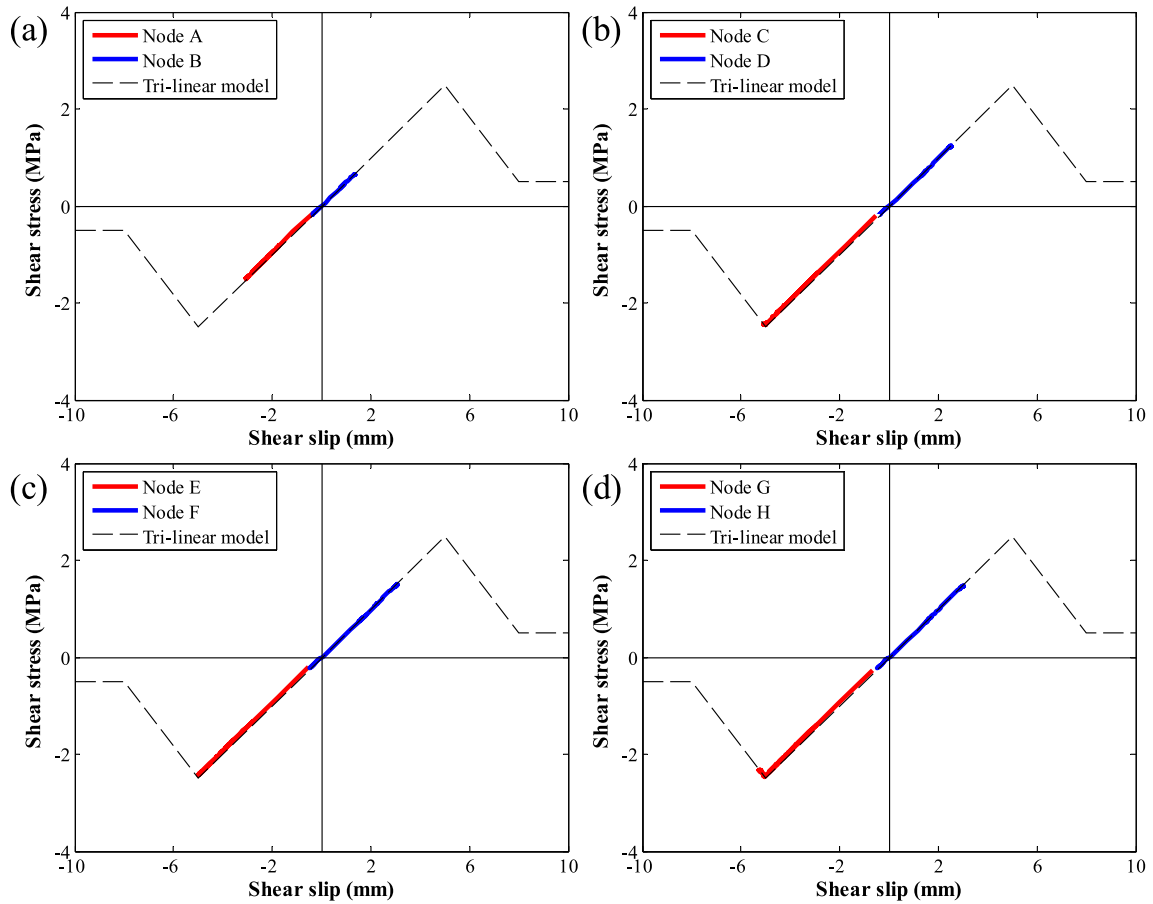


Fig. 13. Development process of the shearing stress at the grout/rock contact surface: (a) Position A, B; (b) Position C, D; (c) Position E, F; (d) Position G, H.

grout/rock contact surface at the internal part was smaller. Following the pre-defined model, much less shearing stress generated and the shearing stress was not able to reach the interfacial shearing stress.

3.3.2. Pillar size

Further work was continued to reveal the pillar size's influence on the anchorage performance of anchor bolts. The boundary condition and the initial setting of this numerical mining stope was same as the description indicated in Sections 3.1 and 3.2.

However, in this sub-section, the pillar size was changed. Specifically, three different pillar size was used: 50 m, 100 m and 150 m (Fig. 14).

As the numerical mining stope finally equilibrated, the tensile force state along anchor bolts is plotted (Fig. 15). It shows that although the pillar size was different, there was not apparent difference on the tensile force state along anchor bolts.

This indicated that at the current situation, a minimum pillar size of 50 m was still able to maintain the stability of the roadway.

3.3.3. Shearing strength of the grout/rock contact surface

Considering that in the underground environment, the shearing strength of the grout/rock contact surface can be adjusted by using high-strength anchorage agent or improving the grouting quality. Therefore, in this sub-section, numerical simulation was performed to reveal the impact of the interfacial shearing strength on the anchorage performance of anchor bolts.

The boundary condition and the initial setting of this numerical mining stope was same as the description indicated in Sections 3.1 and 3.2. However, in this scenario, the shearing strength of the grout/rock contact surface was changed. Specifically, three different values were used: 1 MPa, 2.5 MPa and 4.5 MPa (Fig. 16).

After the numerical mining stope finally equilibrated, the tensile force state in anchor bolts is shown in Fig. 17. Apparently, the shearing strength of the grout/rock contact surface had a marked effect on the uniaxial force of anchor bolts. As the shearing strength of the grout/rock contact surface was 1 MPa, the maximum tensile force in anchor bolts was 47.4 kN. With the increasing shearing strength of the grout/rock contact surface, the maximum tensile force in anchor bolts also ascended. For example, as the shearing strength of the grout/rock contact surface ascended to 4.5 MPa, the maximum tensile force in anchor bolts ascended to 104 kN.

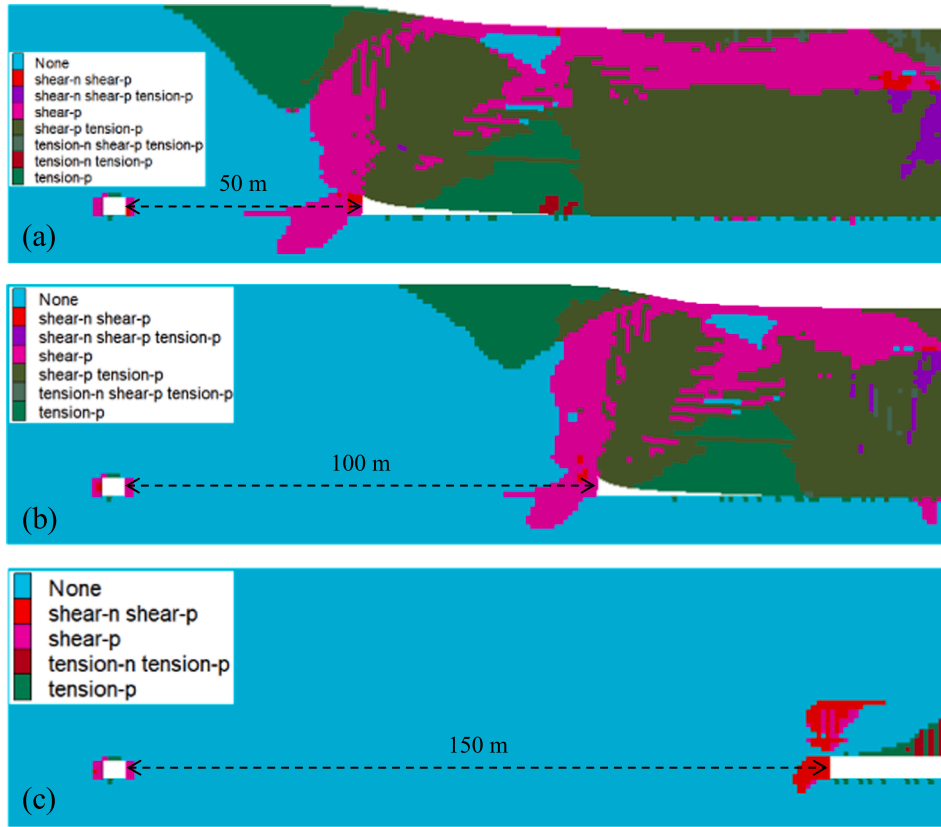


Fig. 14. Plastic zone state: (a) pillar size was 50 m; (b) pillar size was 100 m; (c) pillar size was 150 m.

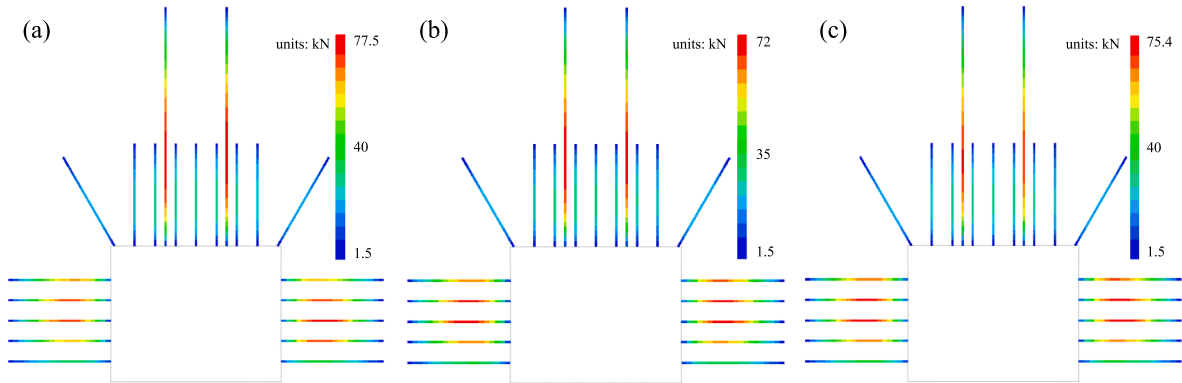


Fig. 15. Tensile force state along anchor bolts: (a) pillar size was 50 m; (b) pillar size was 100 m; (c) pillar size was 150 m.

Increasing of the tensile force in anchor bolts was conducive to preventing the rock deformation around the roadway. Comparing the convergence of the roadways shows that as the shearing strength of the grout/rock contact surface ascended from 1 MPa to 4.5 MPa, the maximum convergence of the roadway decreased. This indicated that to prevent the roadway convergence, increasing the shearing strength of the grout/rock contact surface was an effective method.

4. Recommendation of the future work

This study performed a numerical analysis on the anchorage performance of full-grouted anchor bolts with a revised model. At the current study, all anchor bolts were entirely grouted. However, in coal mines in China, for the rapid tunnelling requirement, the point-anchored method was usually used on anchor bolts. Therefore, to meet with the in-situ scenario, further work will be continued to compare the anchorage performance of point-anchored bolts and full-grouted anchor bolts. Also, further work will be performed to

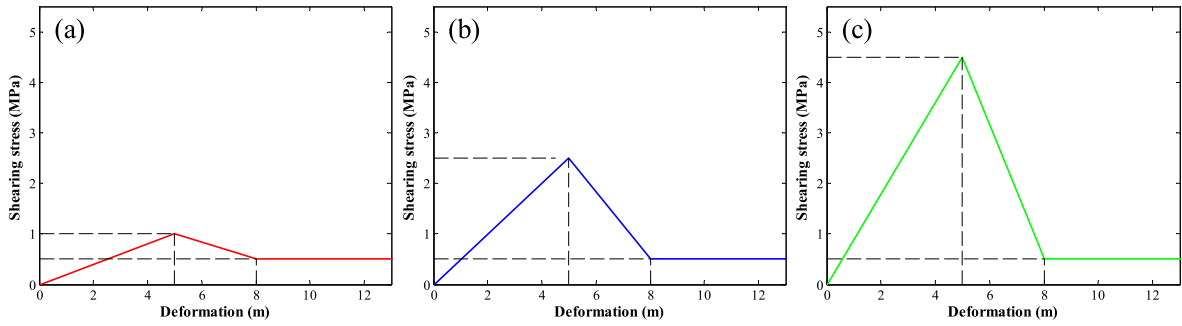


Fig. 16. Inputted shearing stress-deformation relationships: (a) shearing strength was 1 MPa; (b) shearing strength was 2.5 MPa; (c) shearing strength was 4.5 MPa.

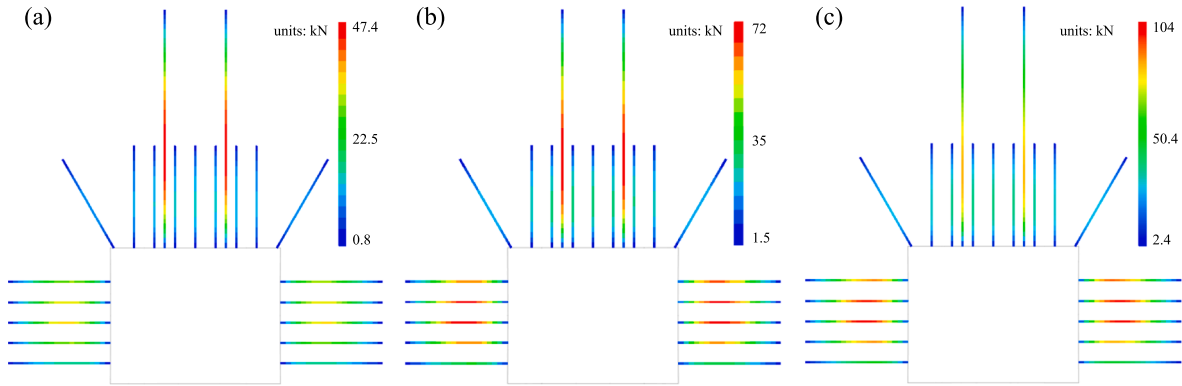


Fig. 17. Tensile force state along anchor bolts: (a) shearing strength was 1 MPa; (b) shearing strength was 2.5 MPa; (c) shearing strength was 4.5 MPa.

reveal the anchorage performance of point-anchored bolts under the in-situ scenario.

5. Conclusions

This study performed a numerical analysis on the anchorage performance of full-grouted anchor bolts with a tri-linear model. It considers the elastic and plastic action of the grout/rock contact surface. Then, it was embedded in the Pile elements in FLAC3D. After that, the anchorage performance of anchor bolts under the in-situ mining condition was studied. The following conclusions were obtained:

- (1) Compared with the elastic-perfectly plastic model in the Pile element, the tri-linear model can consider the post-peak action of the grout/rock contact surface. After it was embedded in the Pile element, bearing capacity of anchor bolts dropped after it reached the maximum. Moreover, the residual bearing capacity of the anchor bolts can be properly simulated.
- (2) The buried depth had a paramount effect on the anchorage performance of full-grouted anchor bolts. With the ascending buried depth, the maximum tensile force in the bolts ascended. Moreover, for the roadway excavated in the coal seam, as the buried depth was small, the roof bolts showed the maximum tensile force. However, as the buried depth was large, the maximum tensile force occurred in the anchor bolts that were fixed in two sides of the roadway.
- (3) The size of the pillar which lies between the retreat working face's stopping line and the roadway, had no obvious effect on the anchorage performance of anchor bolts. Under the current condition, there was a marginal difference on the anchorage performance of anchor bolts as the pillar size ranged between 50 m and 150 m. This indicated that under the current condition, the pillar size of 50 m was sufficient for maintaining the roadway's stability.
- (4) The shearing strength of the grout/rock contact surface directly influenced the anchorage performance of anchor bolts. As the shearing strength of the grout/rock contact surface rose, the maximum tensile force in the anchor bolts ascended. This was conducive to preventing the roadway convergence. The larger the shearing strength of the grout/rock contact surface, the smaller the convergence of the roadway.

Data availability statement

The data were included in this article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data were included in this article.

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